

The Pol-InSAR Course

Part 10: TomoSAR 1

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

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ESAMAAP

EEBIOMASS

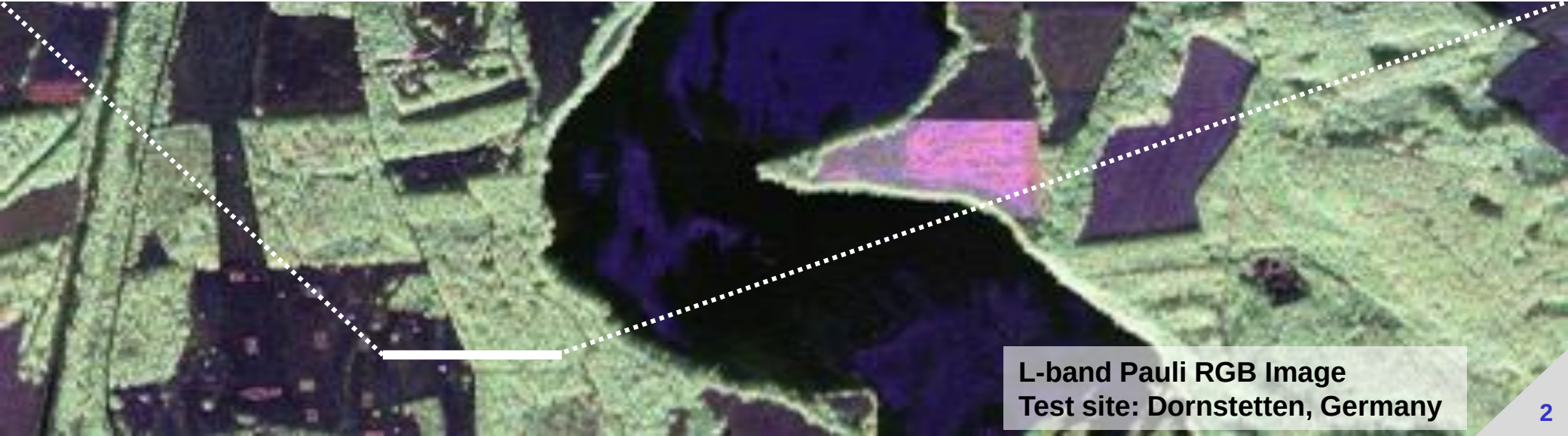
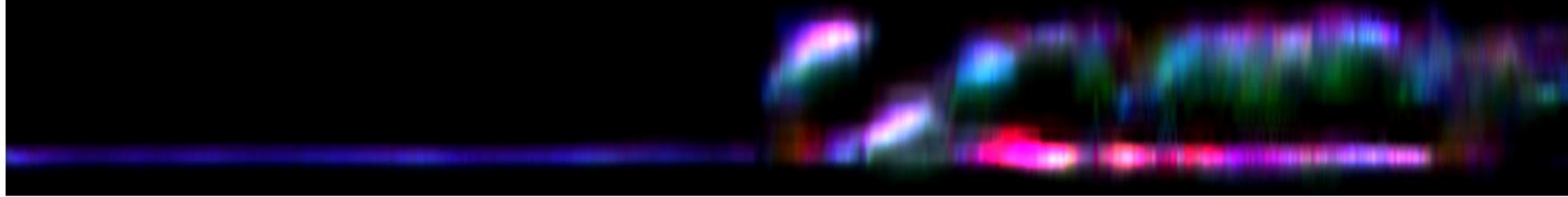


Penetration into vegetation

Vertical Reflectivity Profile (HH)

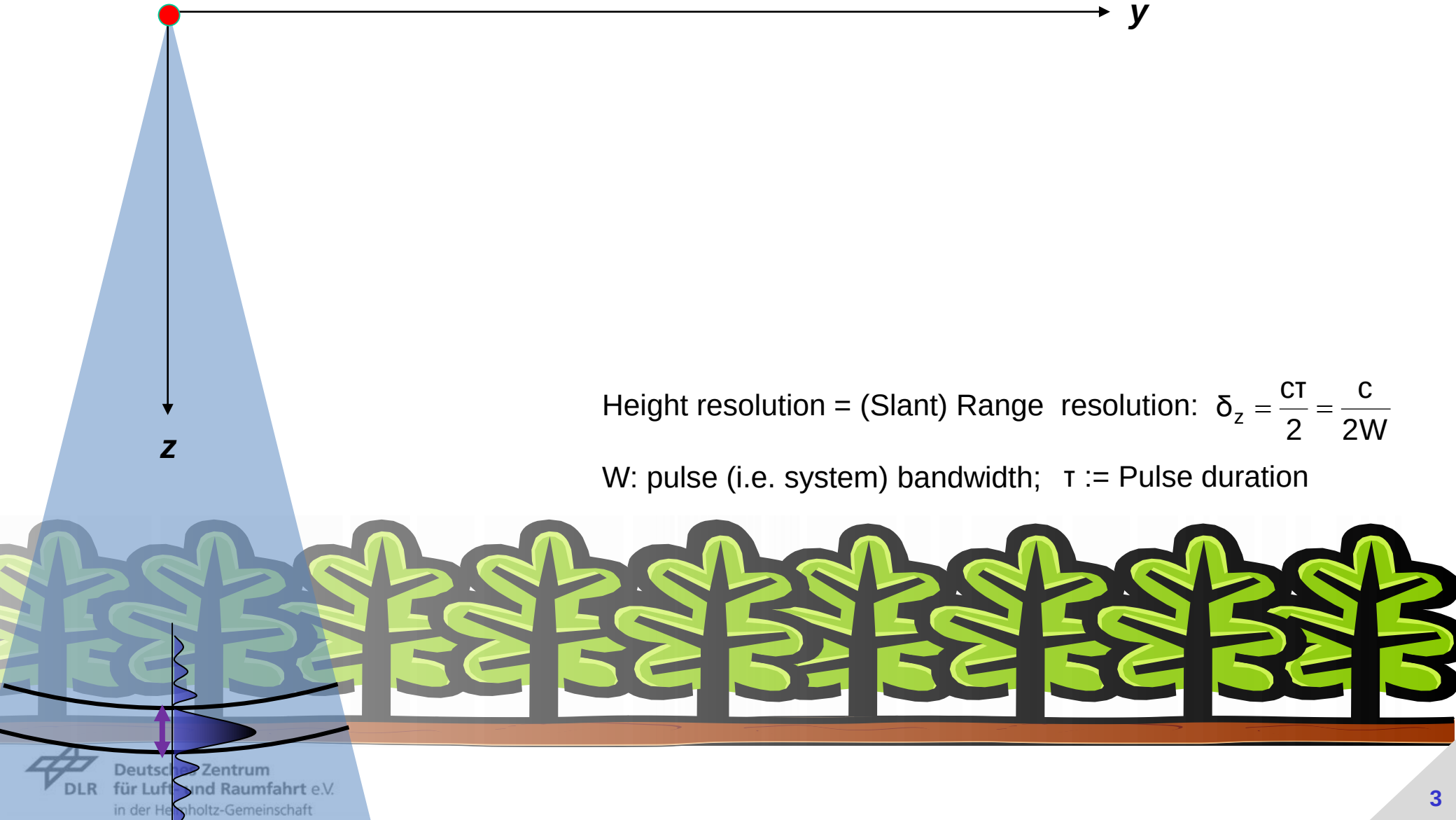


Vertical Reflectivity Profile (Pauli)



L-band Pauli RGB Image
Test site: Dornstetten, Germany

Vertical Resolution: Nadir Looking Case

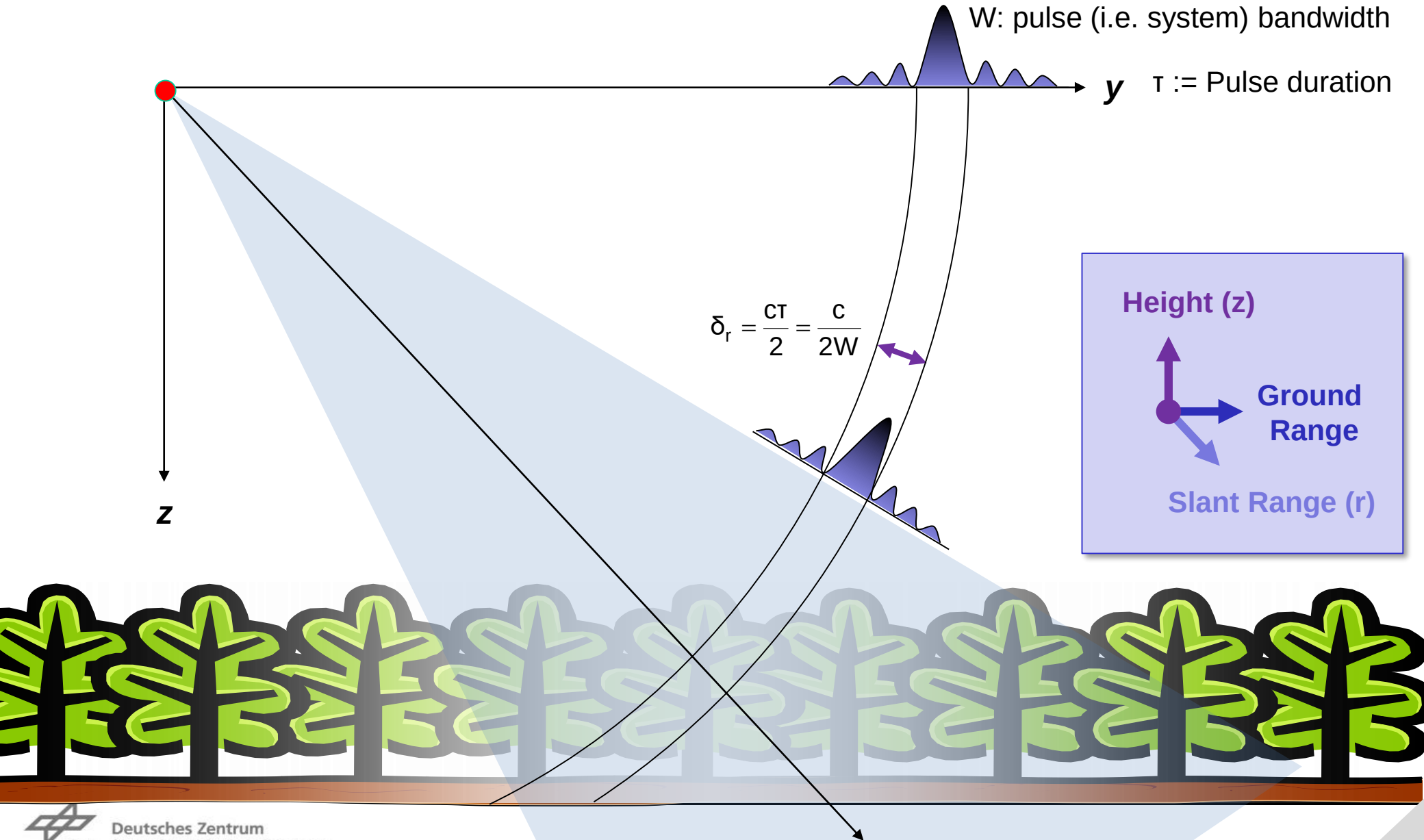


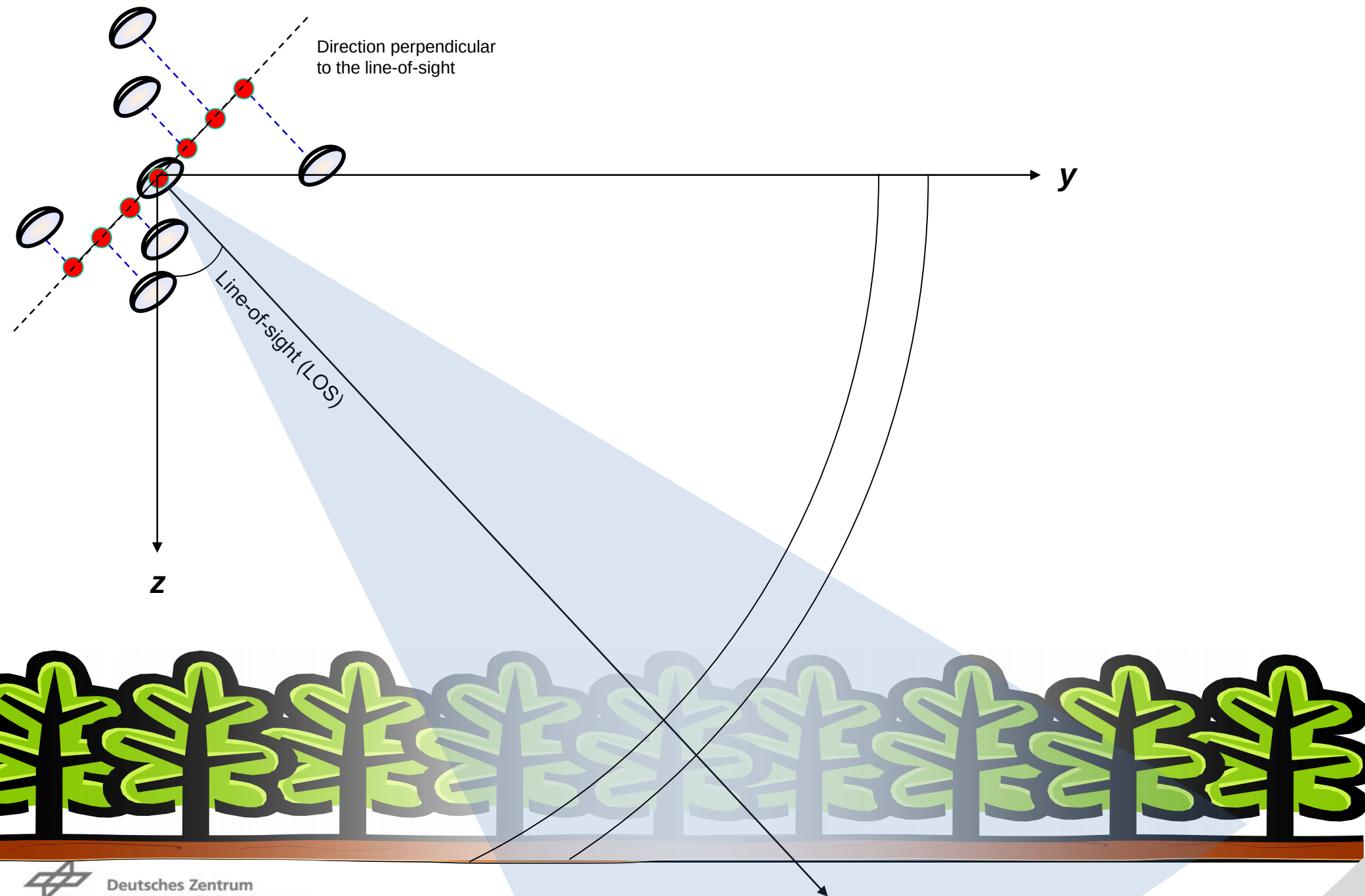
Vertical Resolution: Side Looking Case

(Slant) range resolution: $\delta_r = \frac{c}{2W}$

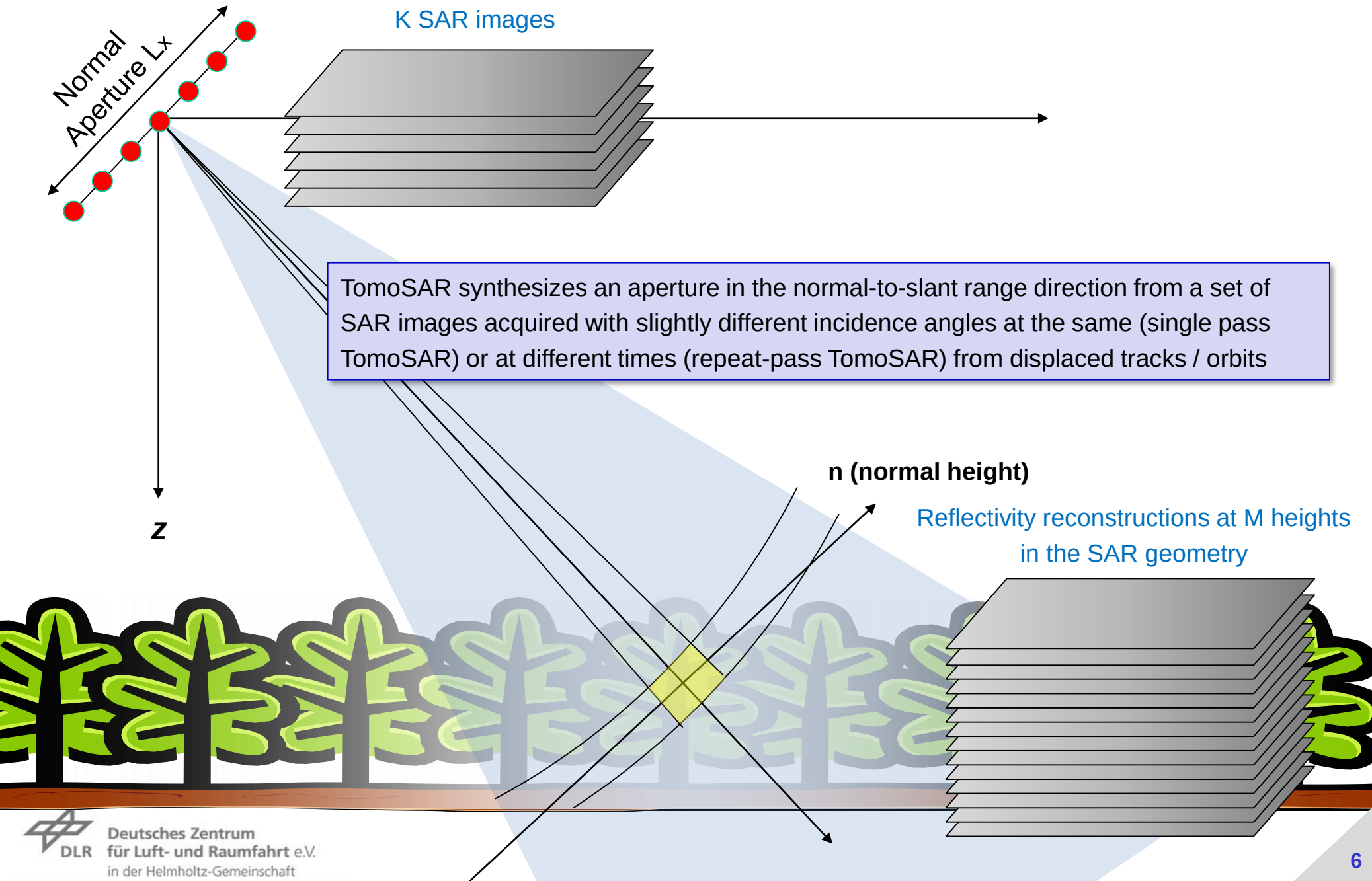
W: pulse (i.e. system) bandwidth

τ := Pulse duration

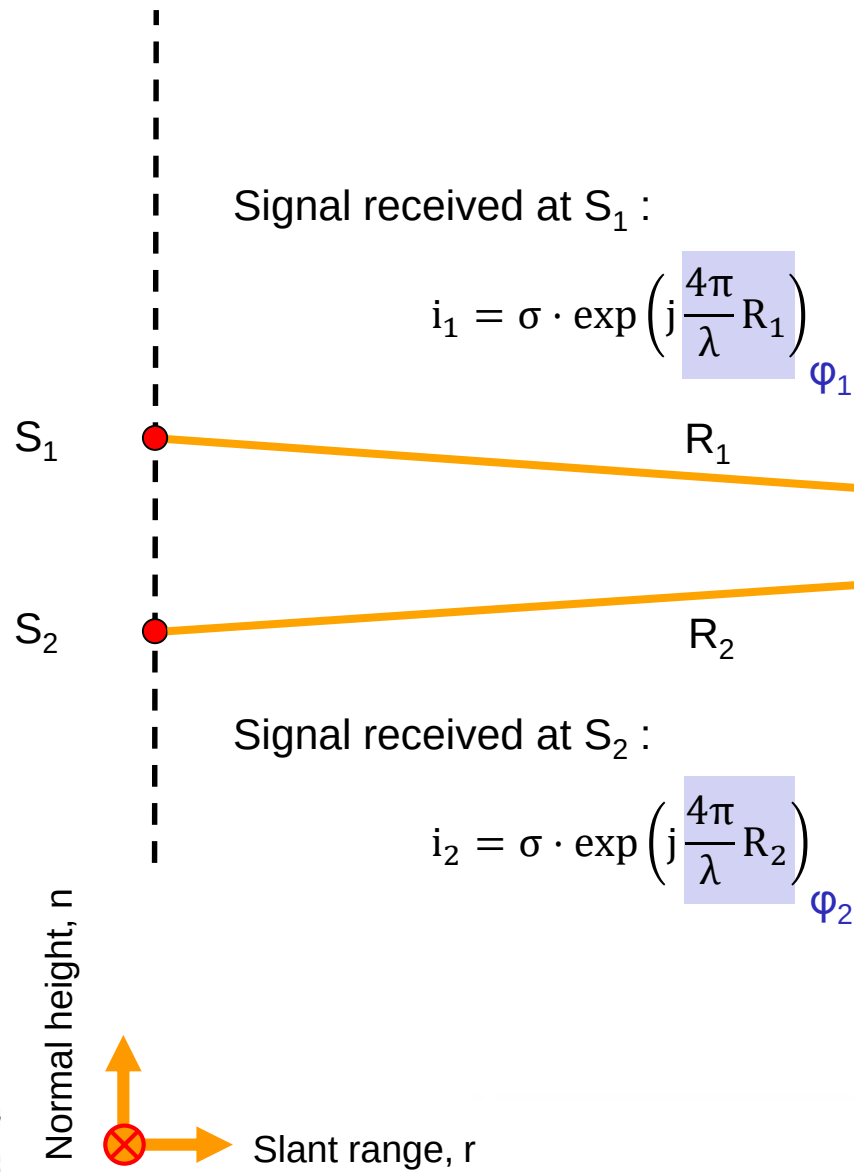




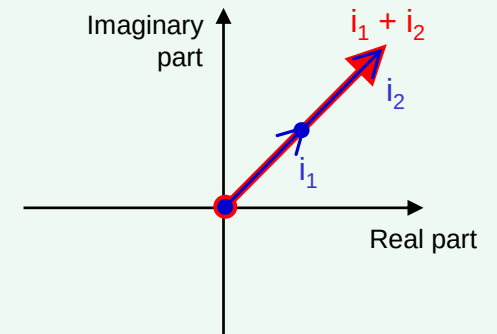
SAR Tomography (TomoSAR)



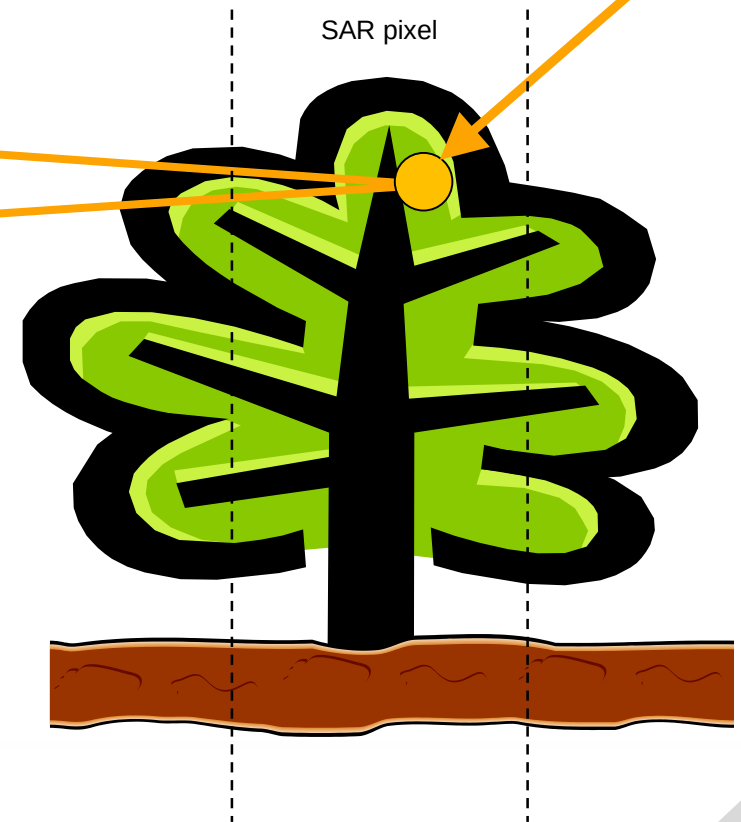
Effect of phase difference



Case:
Phase difference
 $\Delta\varphi = \varphi_2 - \varphi_1 = 0$

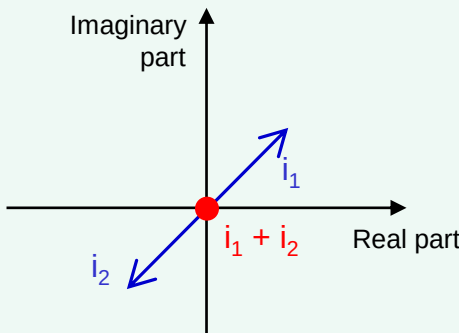


The sum of the two signals (pixel amplitudes) **does not distort** the scattering contribution !!

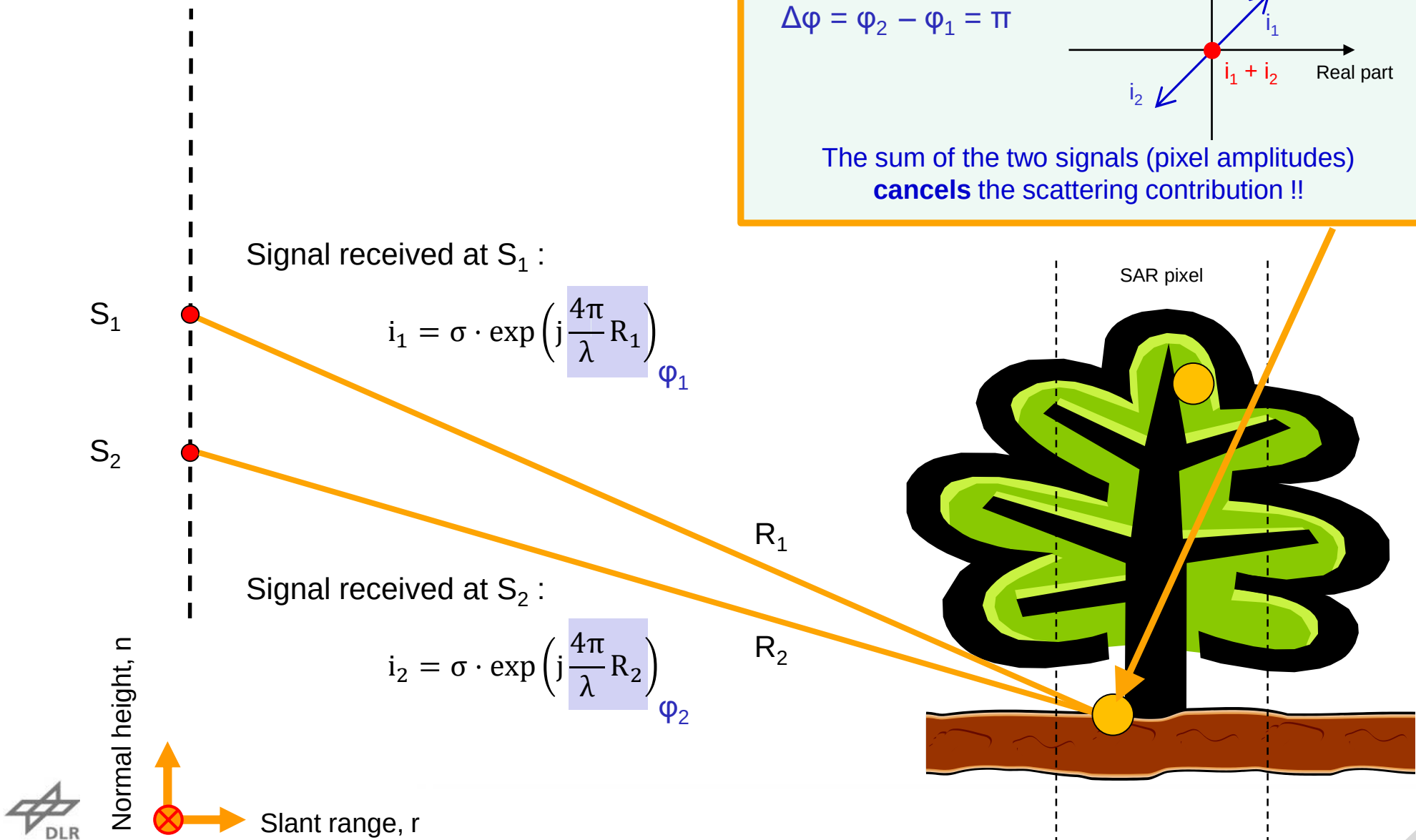


Effect of phase difference

Case:
Phase difference
 $\Delta\varphi = \varphi_2 - \varphi_1 = \pi$



The sum of the two signals (pixel amplitudes)
cancels the scattering contribution !!



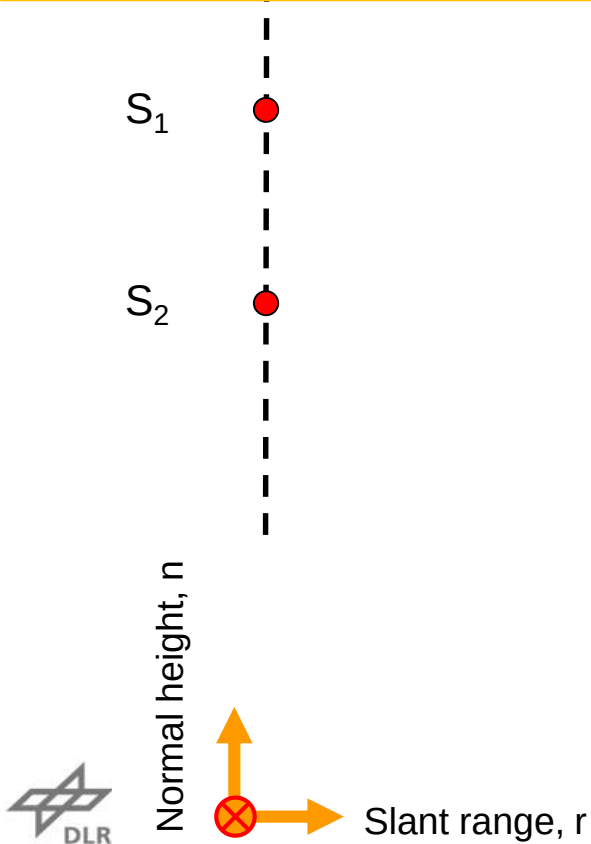
Complex reflectivity reconstruction

For a point at generic slant range-normal height (r, n) :

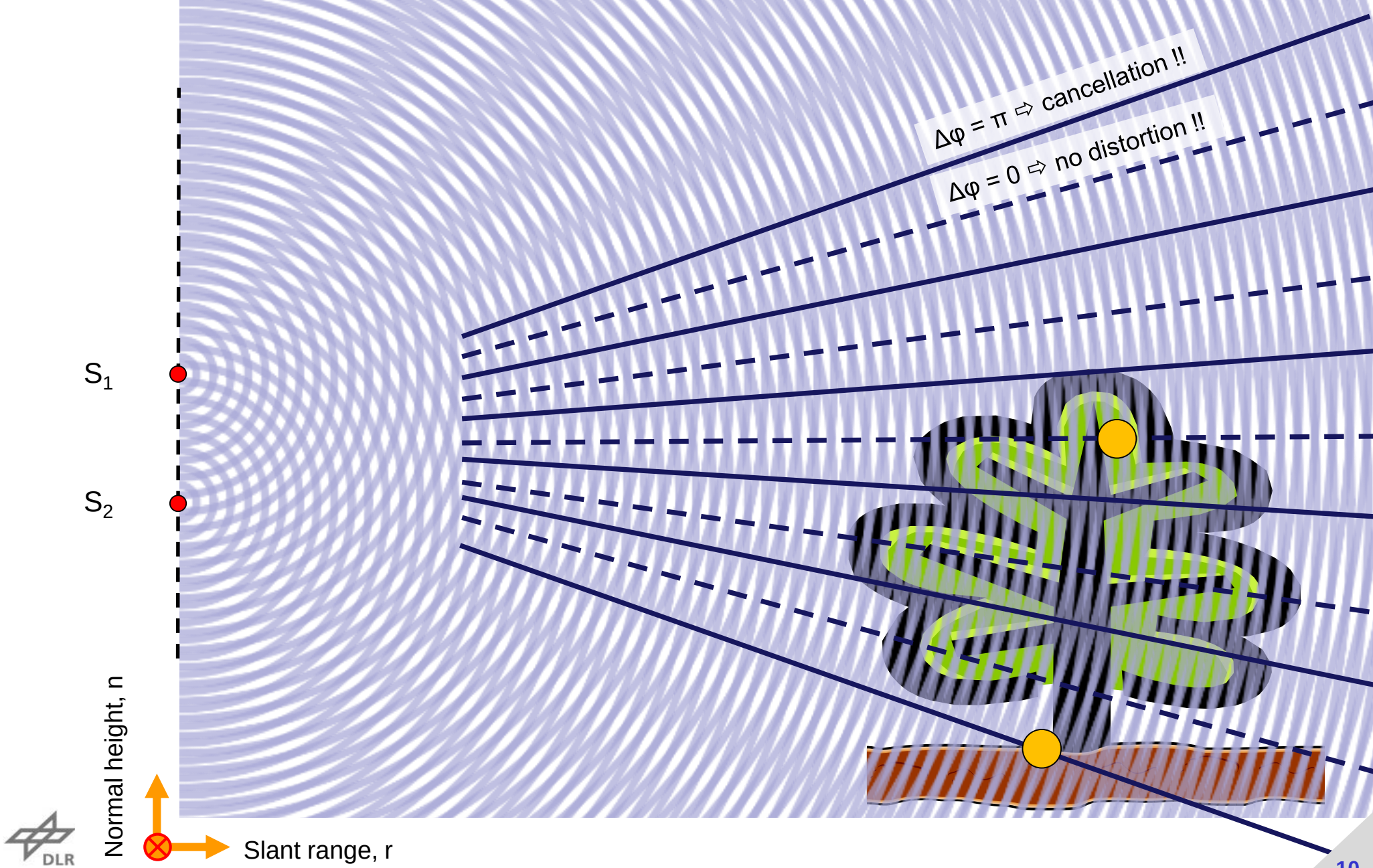
$$i_1(r, n) + i_2(r, n) = \sigma(r, n) \cdot [\exp(j\varphi_1(r, n)) + \exp(j\varphi_2(r, n))] \propto \sigma(r, n) \cdot [1 + \exp(j\Delta\varphi(r, n))]$$

Hence, the capability to cancel or leave undistorted scattering contributions depends on the function:

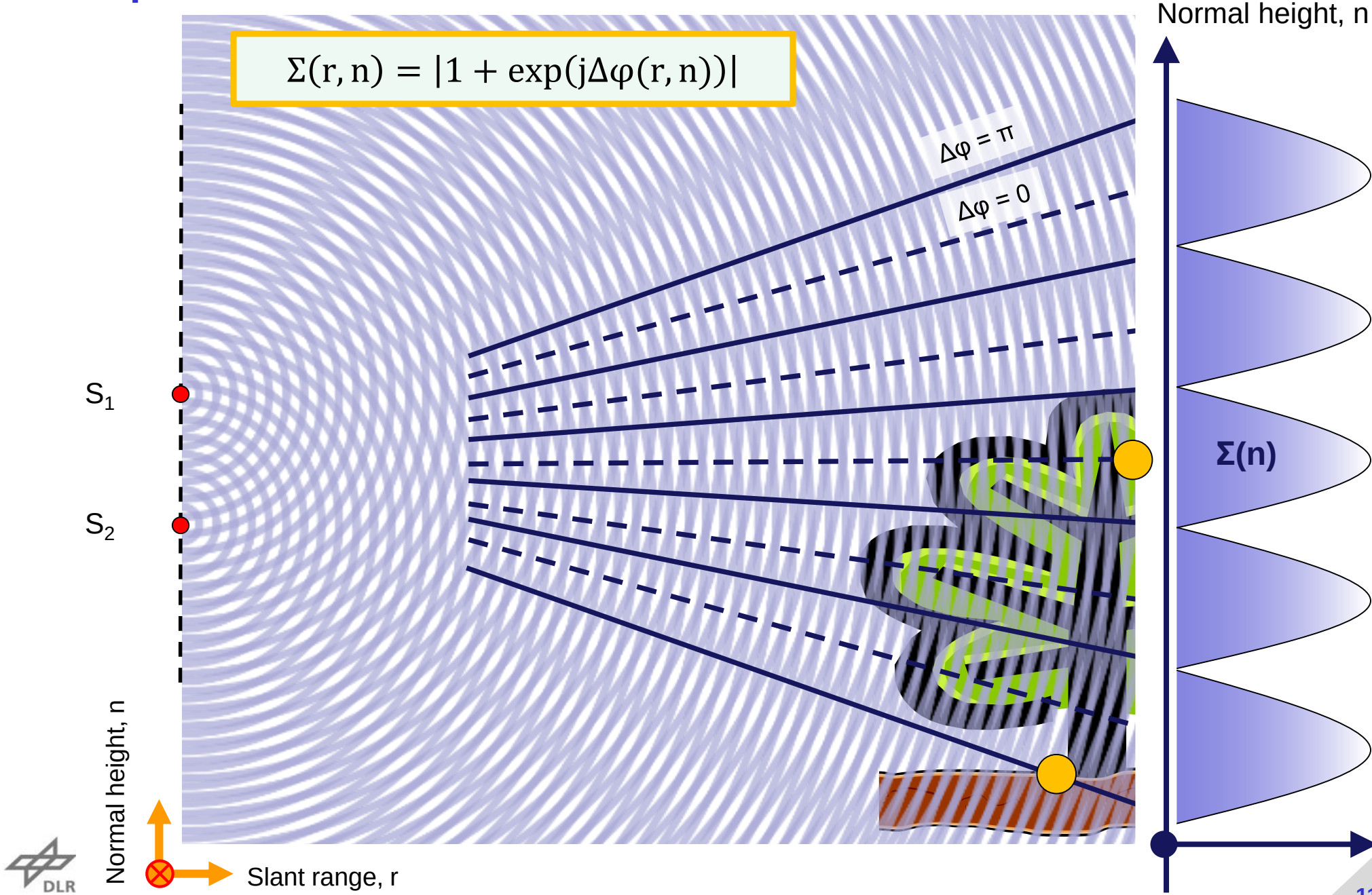
$$\Sigma(r, n) = |1 + \exp(j\Delta\varphi(r, n))|$$



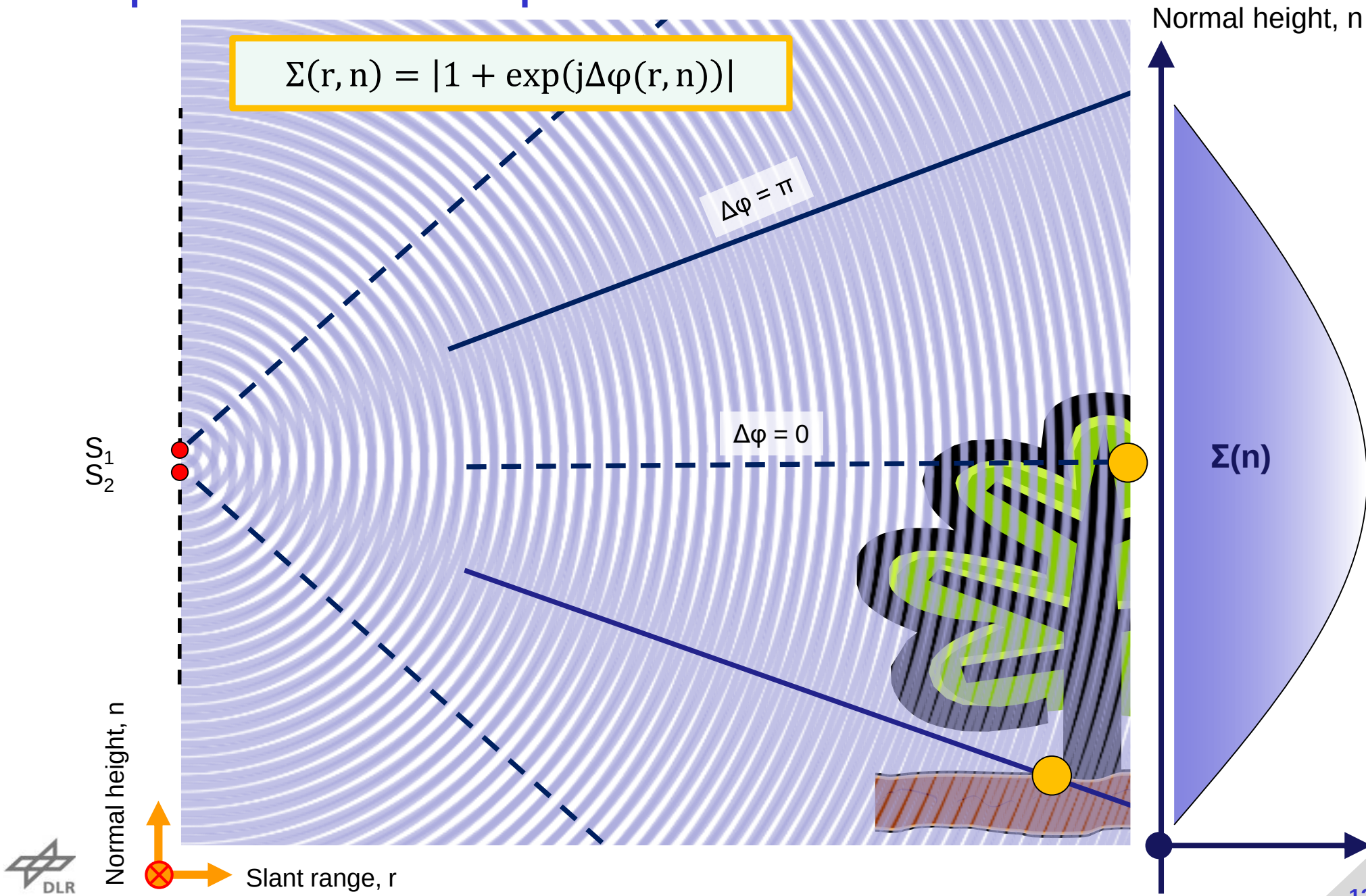
Phase patterns and cancellation



Phase patterns and cancellation



Phase patterns: Shorter displacements



The TomoSAR case

$$\Sigma(r, n) = |1 + \exp(j\Delta\varphi_{21}(r, n)) + \cdots + \exp(j\Delta\varphi_{K1}(r, n))|$$

Phase difference
between S_2 and S_1

Phase difference
between S_K and S_1

TomoSAR (normal) aperture

L_x

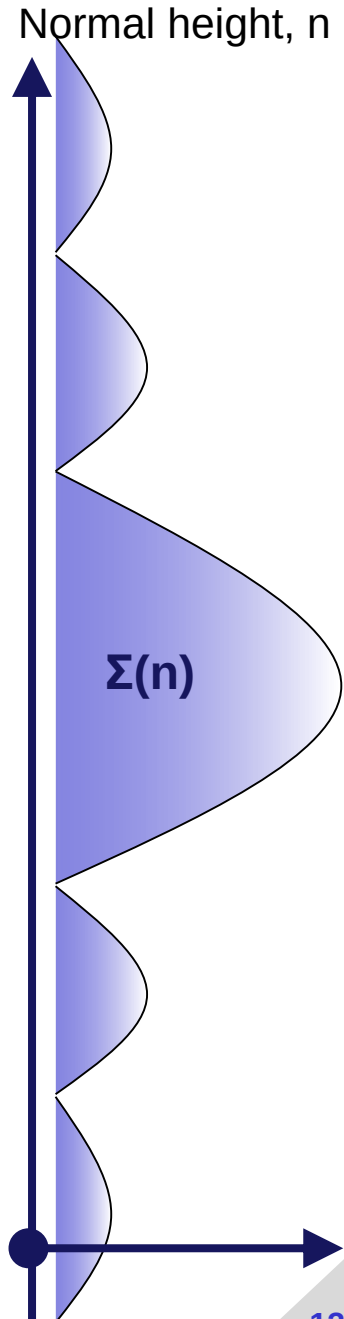
S_1
 S_2

S_K

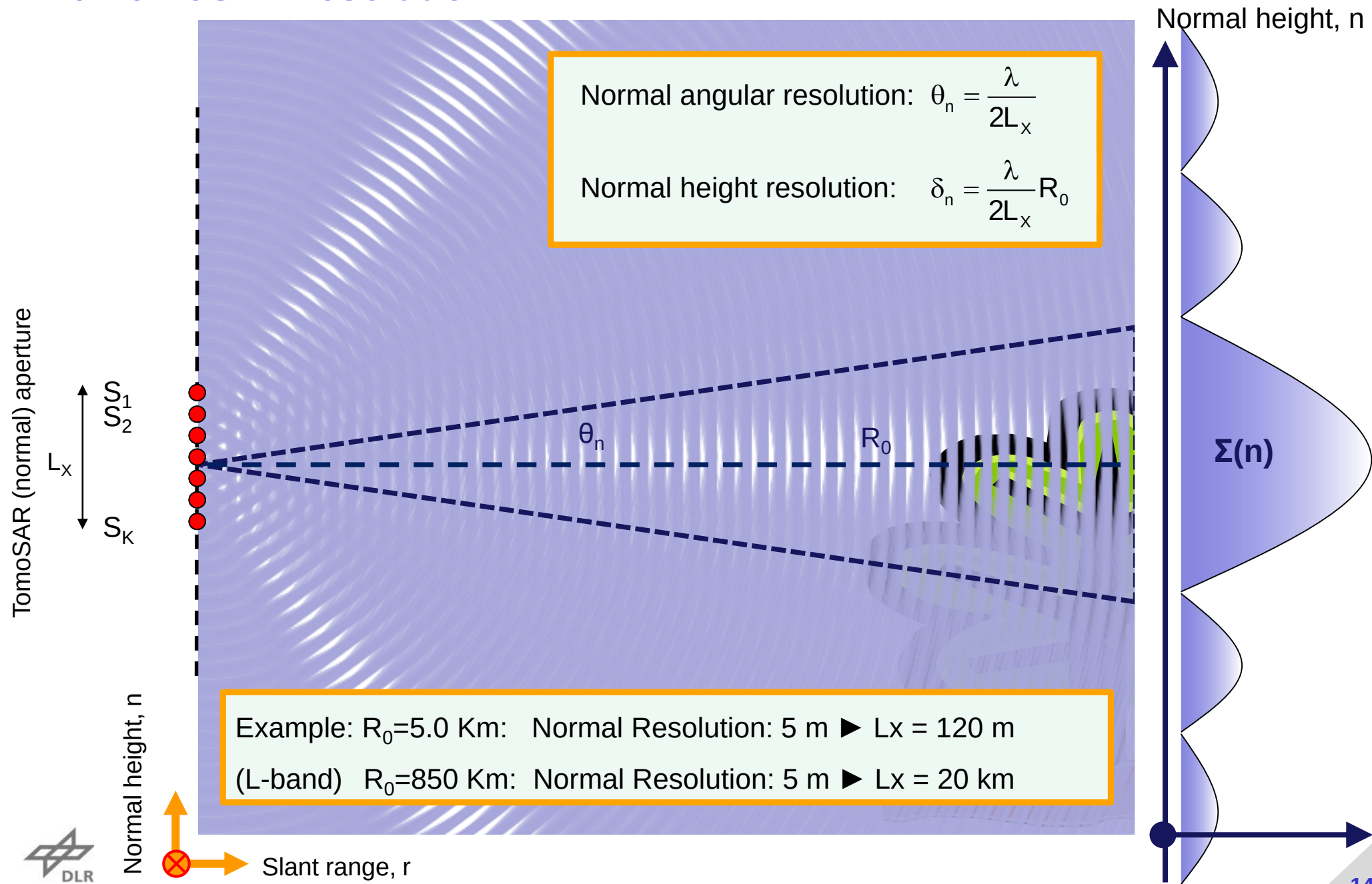
Normal height, n

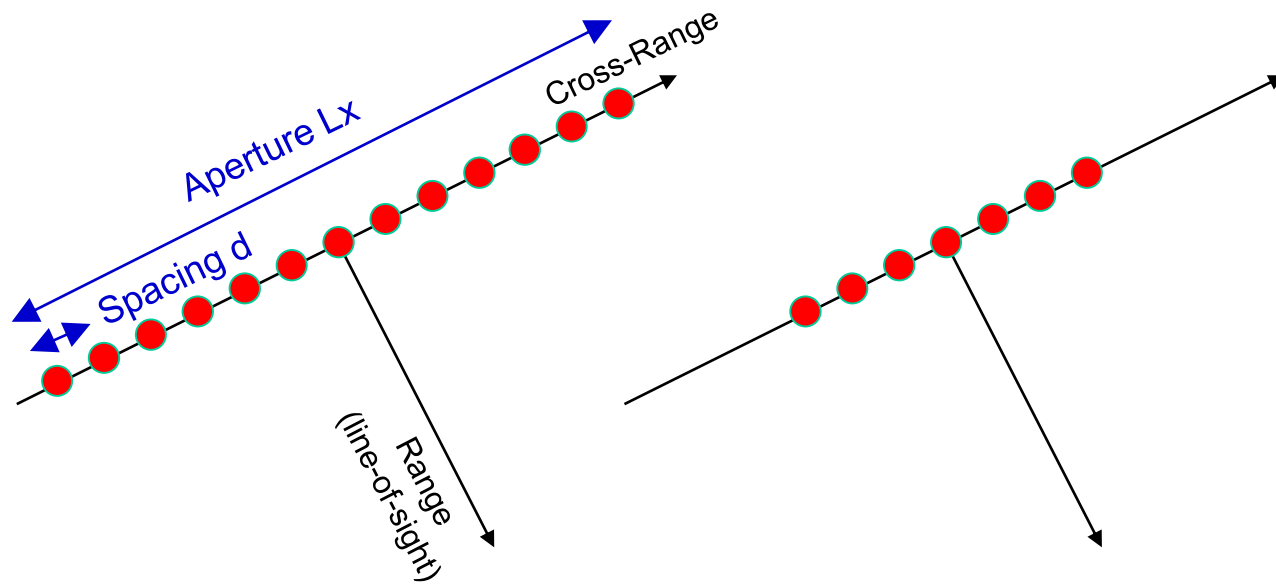


Slant range, r



The TomoSAR resolution

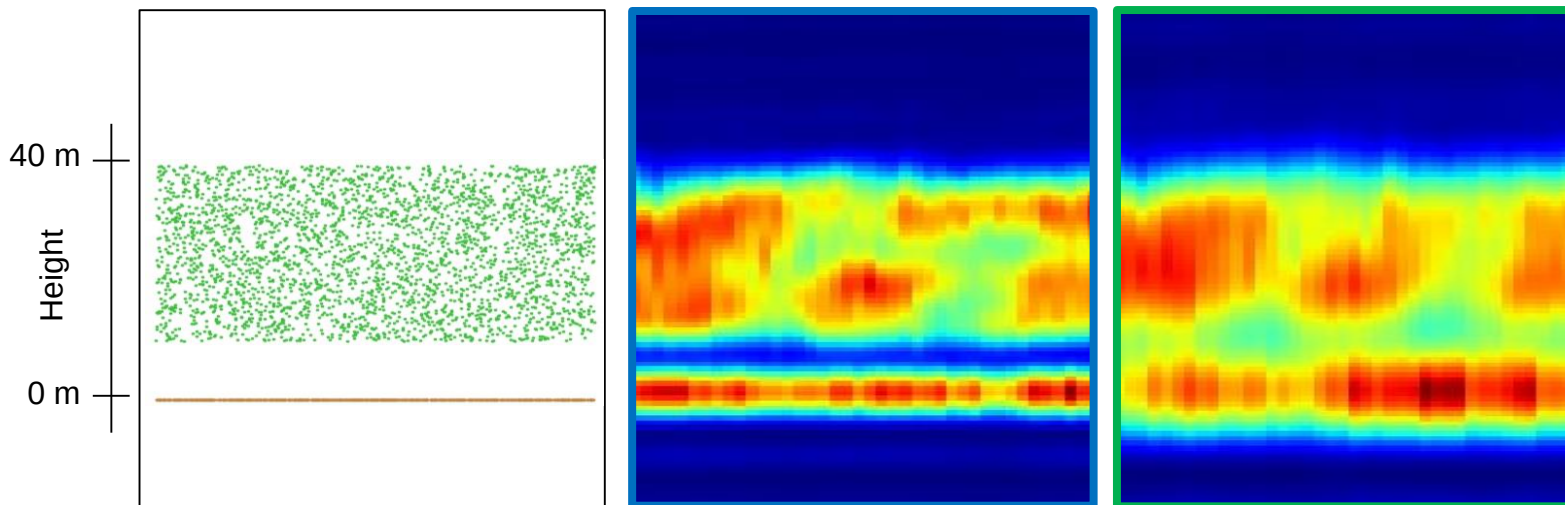




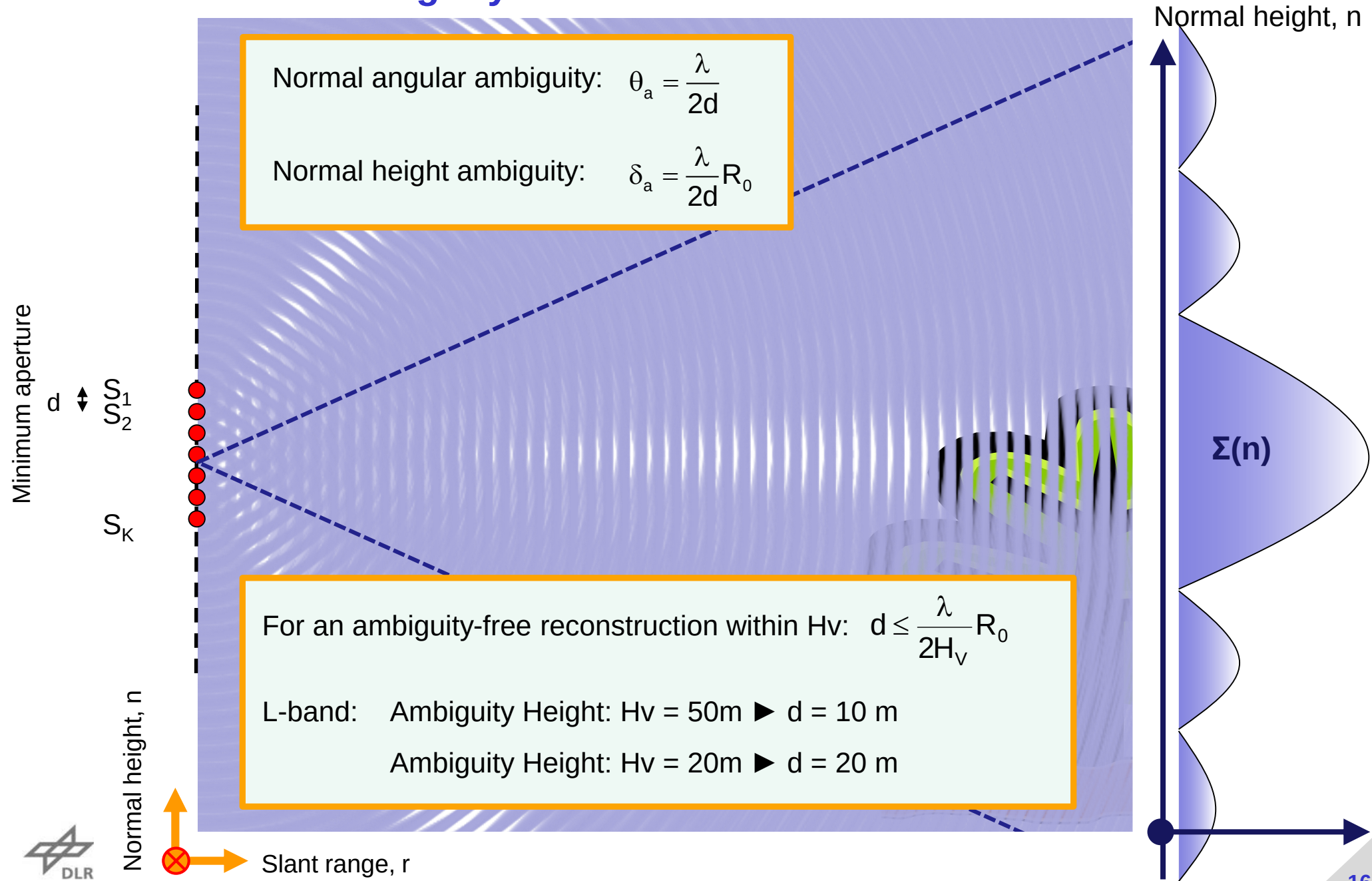
Simulated scene

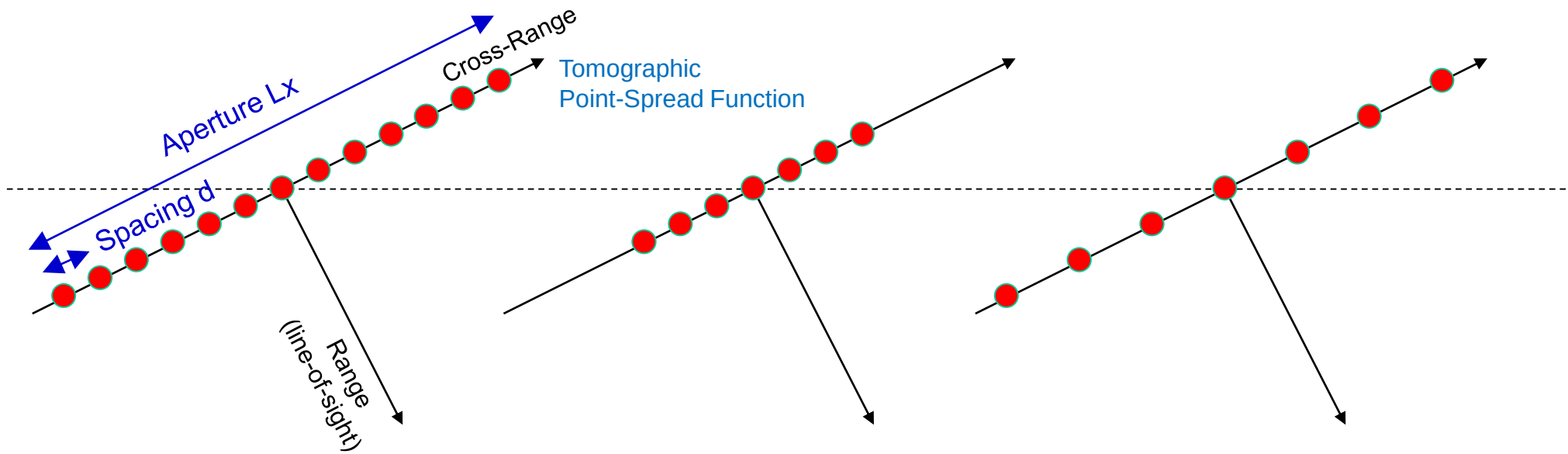
13 uniform tracks
Aperture = L_x
Spacing = d

7 uniform tracks
Aperture = $L_x / 2$
Spacing = d



The TomoSAR ambiguity



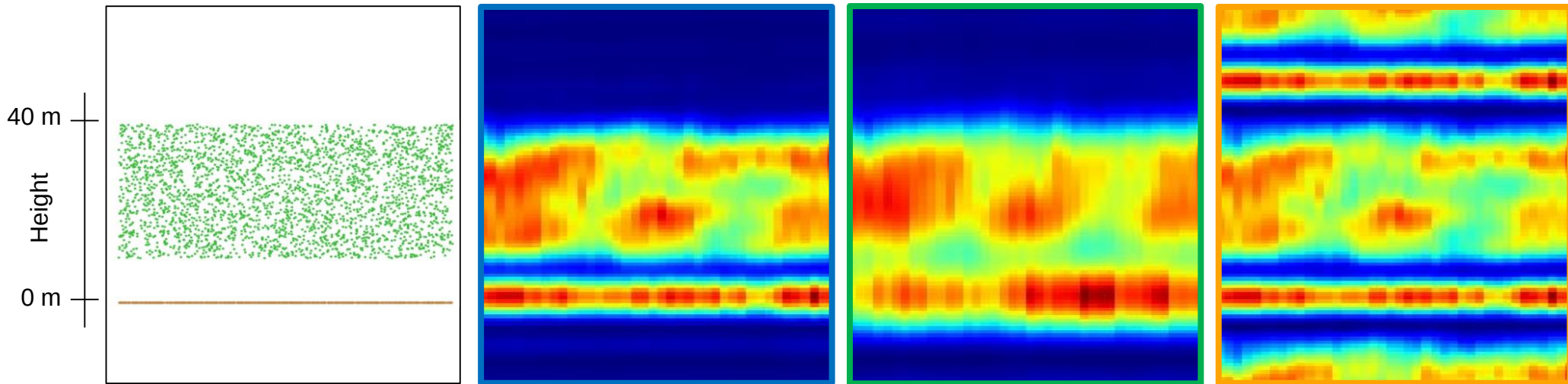


Simulated scene

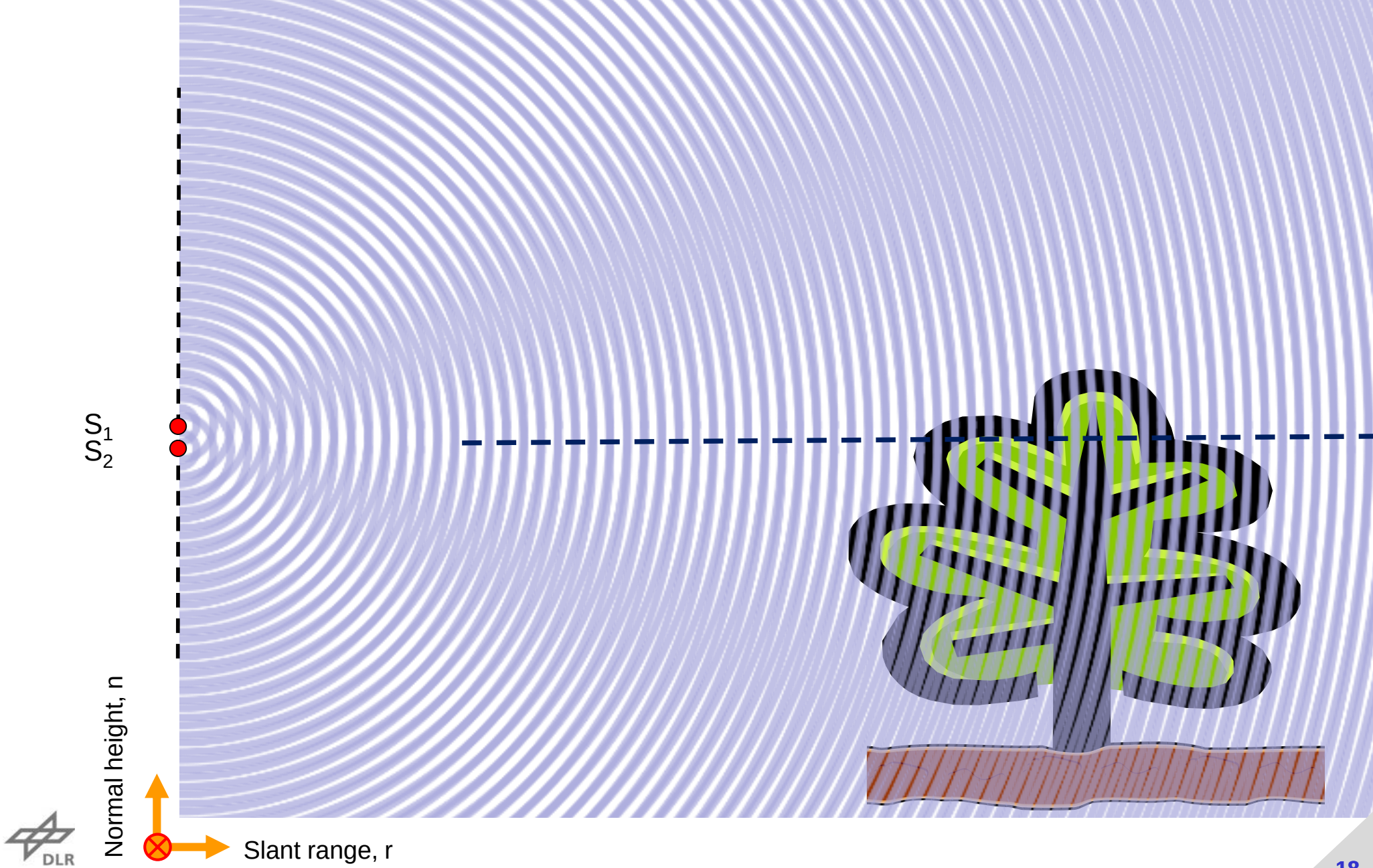
13 uniform tracks
Aperture = L_x
Spacing = d

7 uniform tracks
Aperture = $L_x / 2$
Spacing = d

7 uniform tracks
Aperture = L_x
Spacing = $2d$

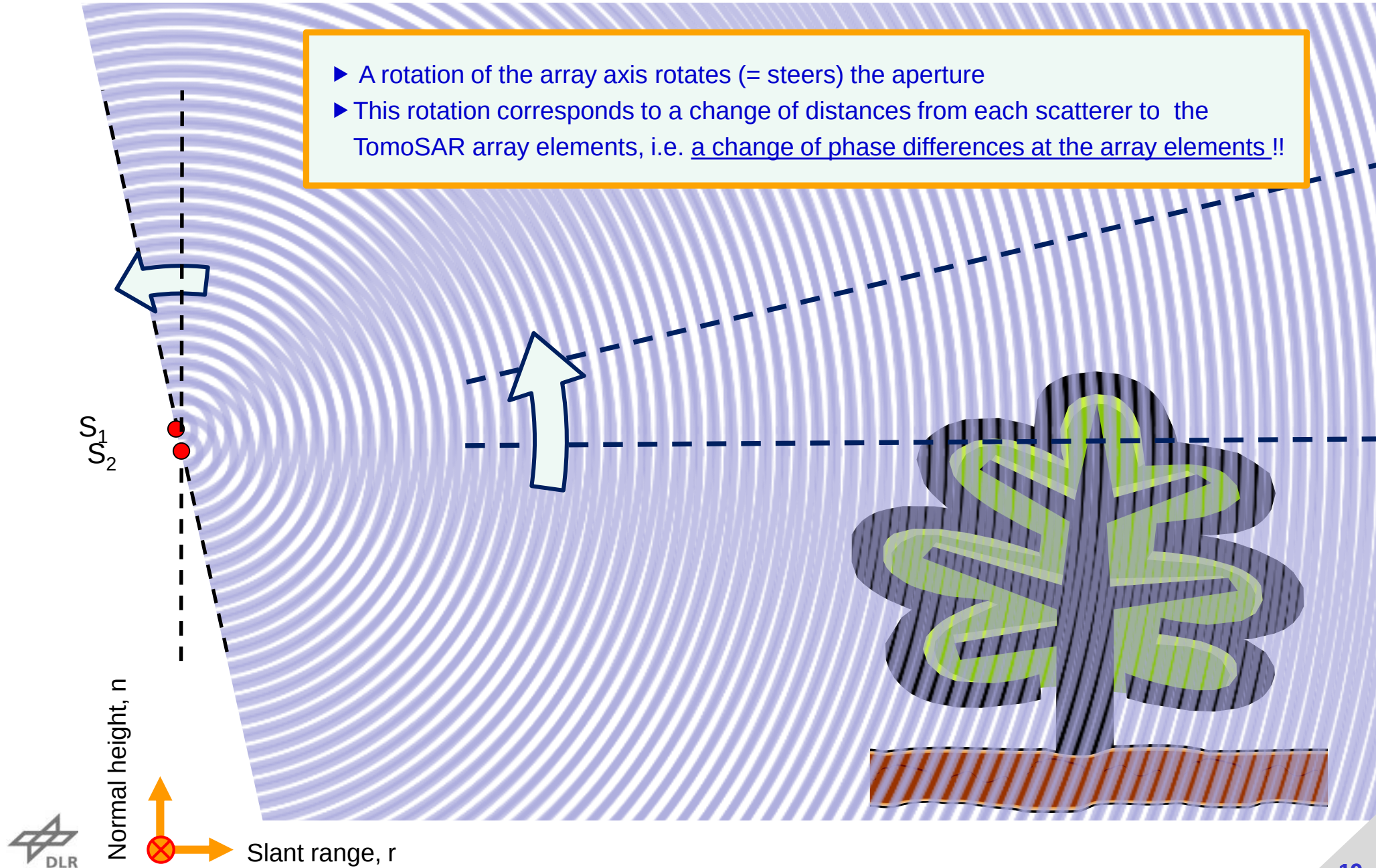


Complex reflectivity reconstruction

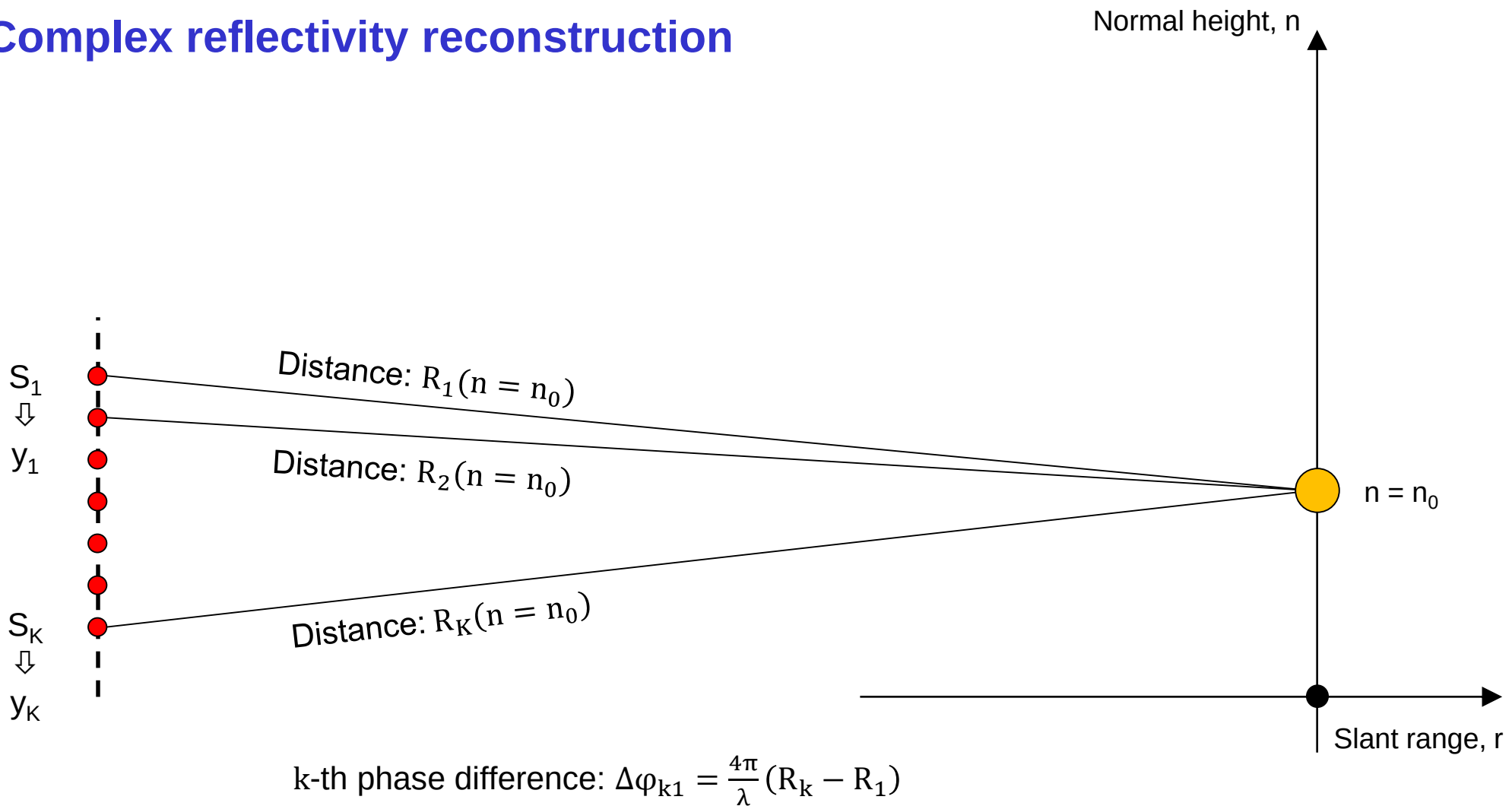


Complex reflectivity reconstruction

- ▶ A rotation of the array axis rotates (= steers) the aperture
- ▶ This rotation corresponds to a change of distances from each scatterer to the TomoSAR array elements, i.e. a change of phase differences at the array elements !!



Complex reflectivity reconstruction



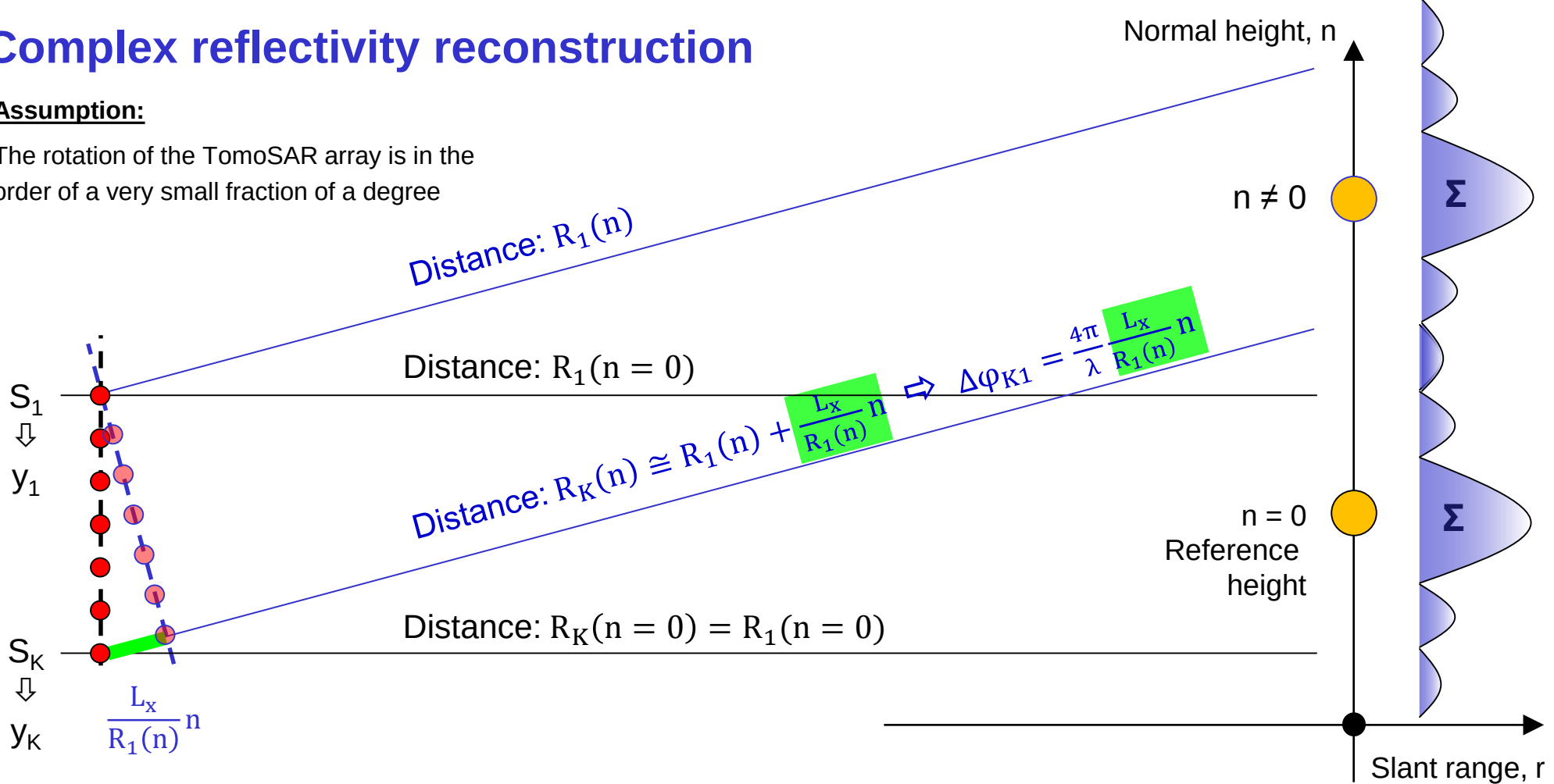
Preliminary processing step: compensate the phase differences of all the acquisitions (TomoSAR array elements) with respect to a reference one for a fixed height

► This fixed height will be the zero-height of the reconstruction !

Complex reflectivity reconstruction

Assumption:

The rotation of the TomoSAR array is in the order of a very small fraction of a degree



Reconstructed Complex Reflectivity

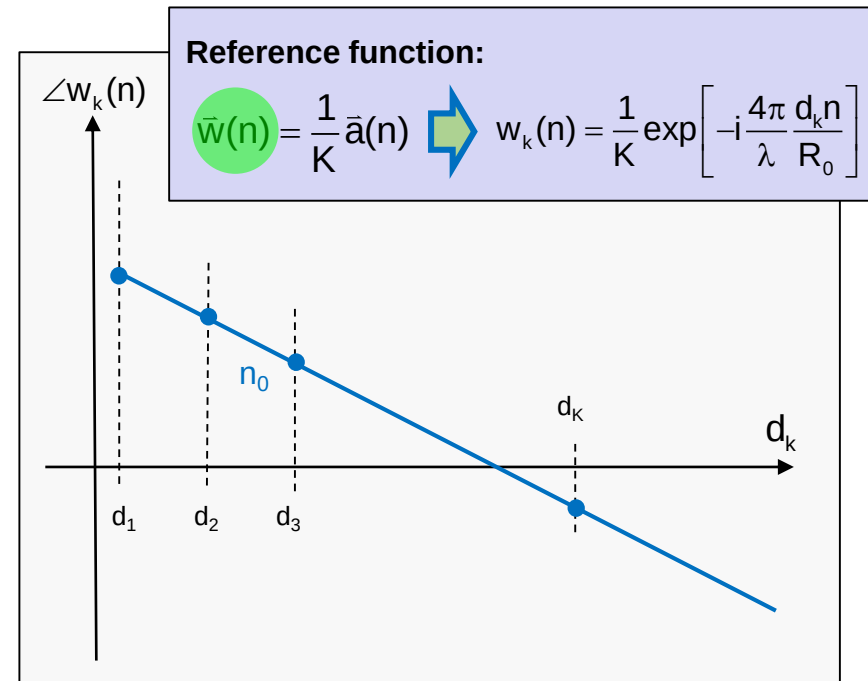
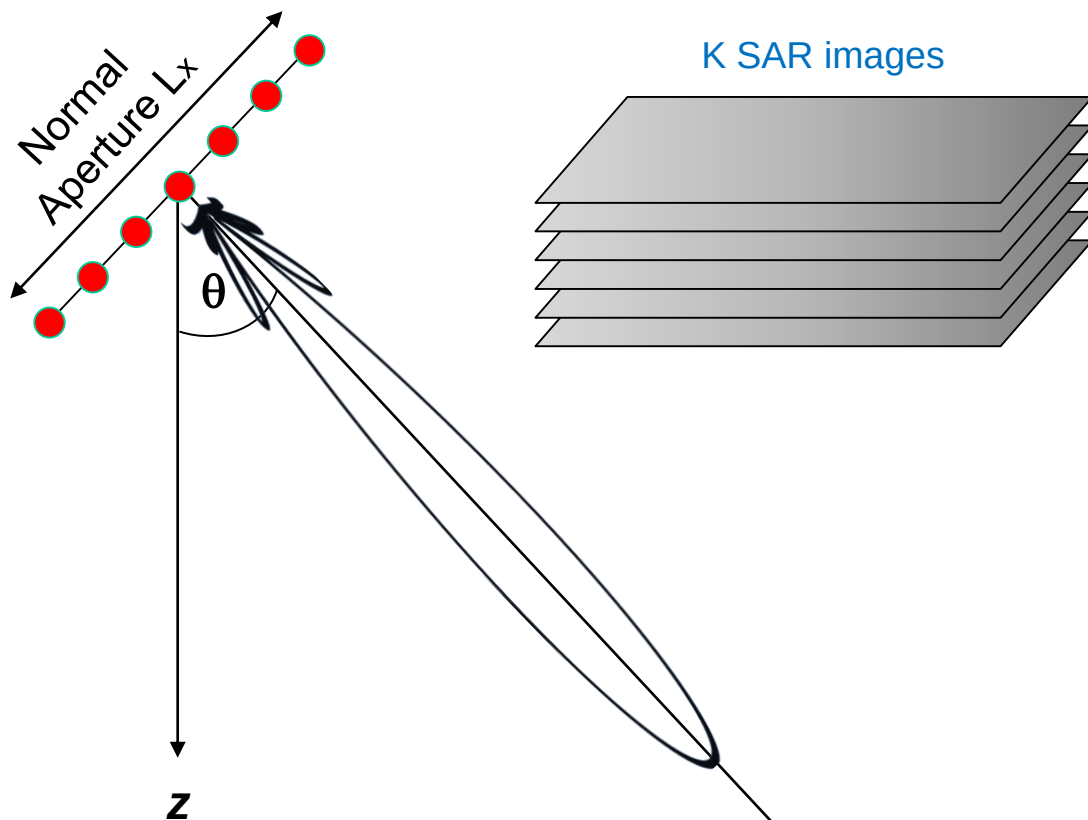
$$\tilde{P}(n) = \frac{1}{K} \left[\exp\left(j \frac{4\pi}{\lambda} \frac{d_1}{R_0} n\right) \cdot y_1 + \dots + \exp\left(j \frac{4\pi}{\lambda} \frac{d_k}{R_0} n\right) \cdot y_k + \dots + \exp\left(j \frac{4\pi}{\lambda} \frac{d_K}{R_0} n\right) \cdot y_K \right]$$

$$\tilde{P}(n) = w_1^* \cdot y_1 + \dots + w_k^* \cdot y_k + \dots + w_K^* \cdot y_K$$

$$w_k = \frac{1}{K} \exp\left(-j \frac{4\pi}{\lambda} \frac{d_k}{R_0} n\right) = \frac{1}{K} \exp[j \cdot \Delta\phi_{k1}(n)] = \frac{1}{K} a_k$$

k-th element of the "steering vector" $\vec{a}(n)$

Complex reflectivity reconstruction



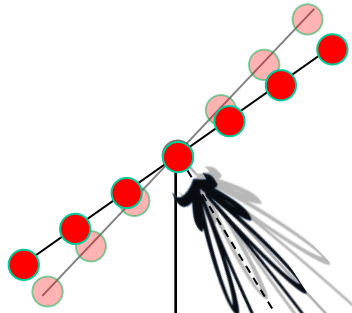
Reflectivity reconstruction:

$$\tilde{P}(n) = \bar{w}^{*T}(n) \cdot \bar{y}$$

Data vector: (each pixel)

$$\bar{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_K \end{bmatrix}$$

Complex reflectivity reconstruction



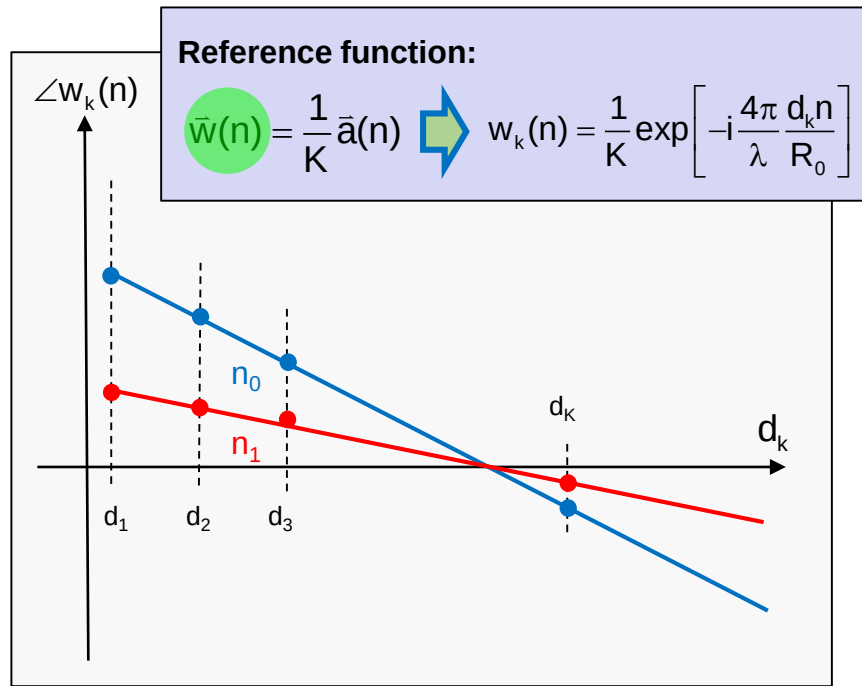
K SAR images



z

Data vector:
(each pixel)

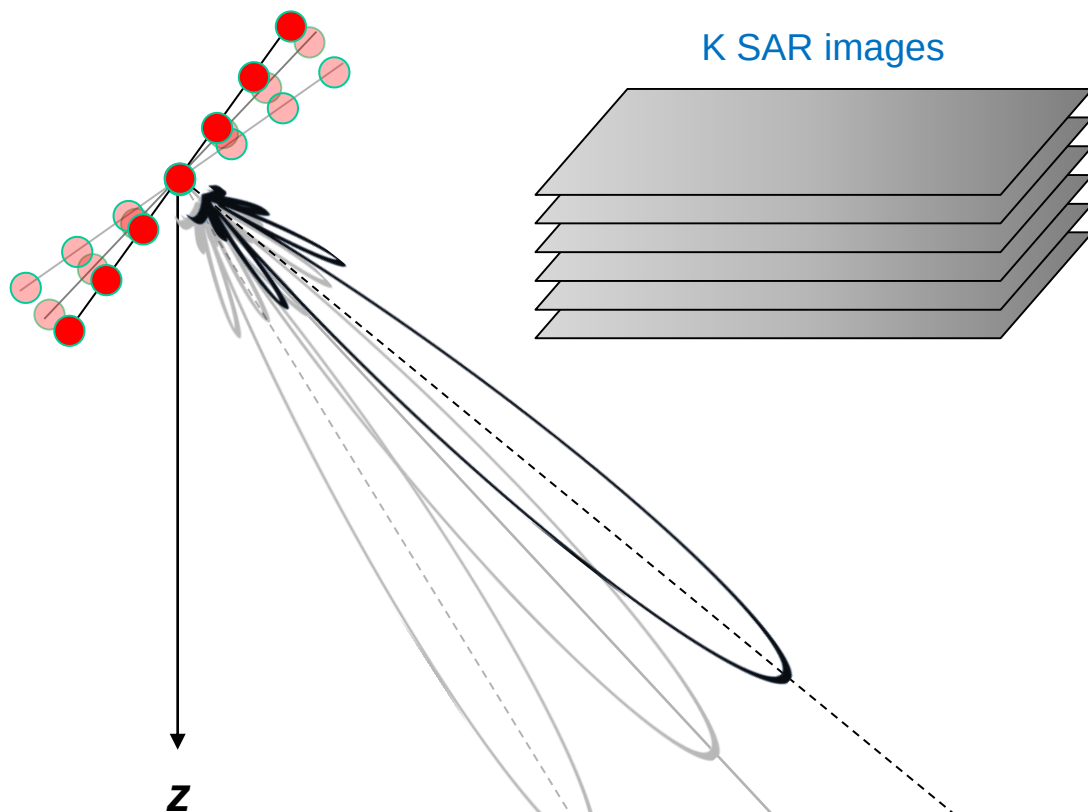
$$\bar{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_K \end{bmatrix}$$



Reflectivity reconstruction:

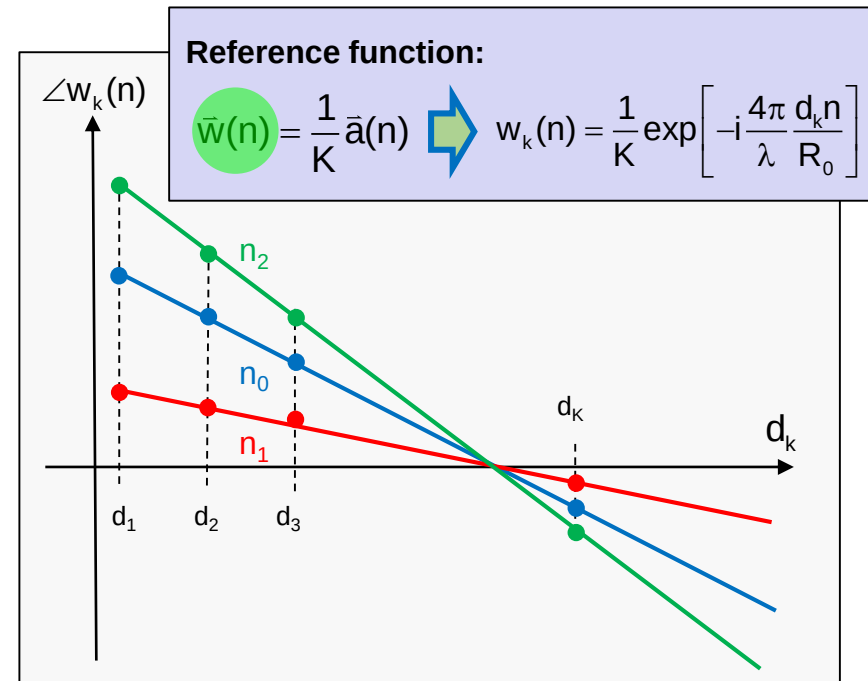
$$\tilde{P}(n) = \bar{w}^{*T}(n) \cdot \bar{y}$$

Complex reflectivity reconstruction



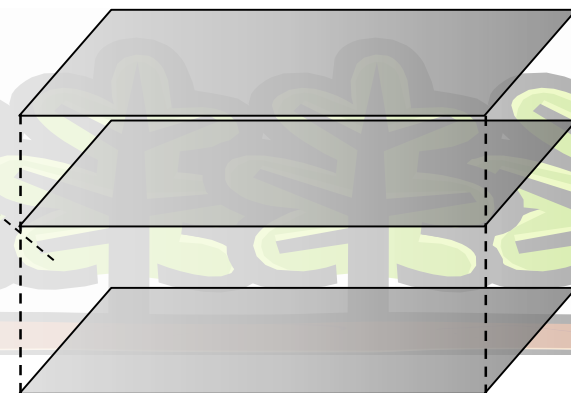
Data vector: (each pixel)

$$\tilde{\mathbf{y}} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_K \end{bmatrix}$$

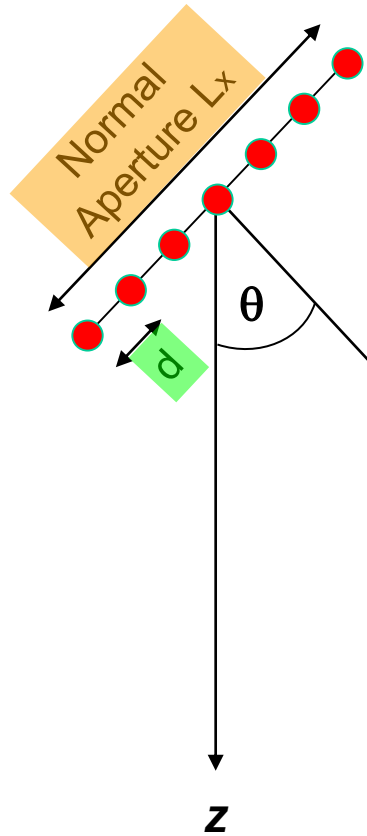


Reflectivity reconstruction:

$$\tilde{P}(n) = \bar{w}^{*T}(n) \cdot \tilde{\mathbf{y}}$$



To summarize



Sampling condition 4 **ambiguity-free** reconstruction within H_V : $d \leq \frac{\lambda}{2H_V} R_0$

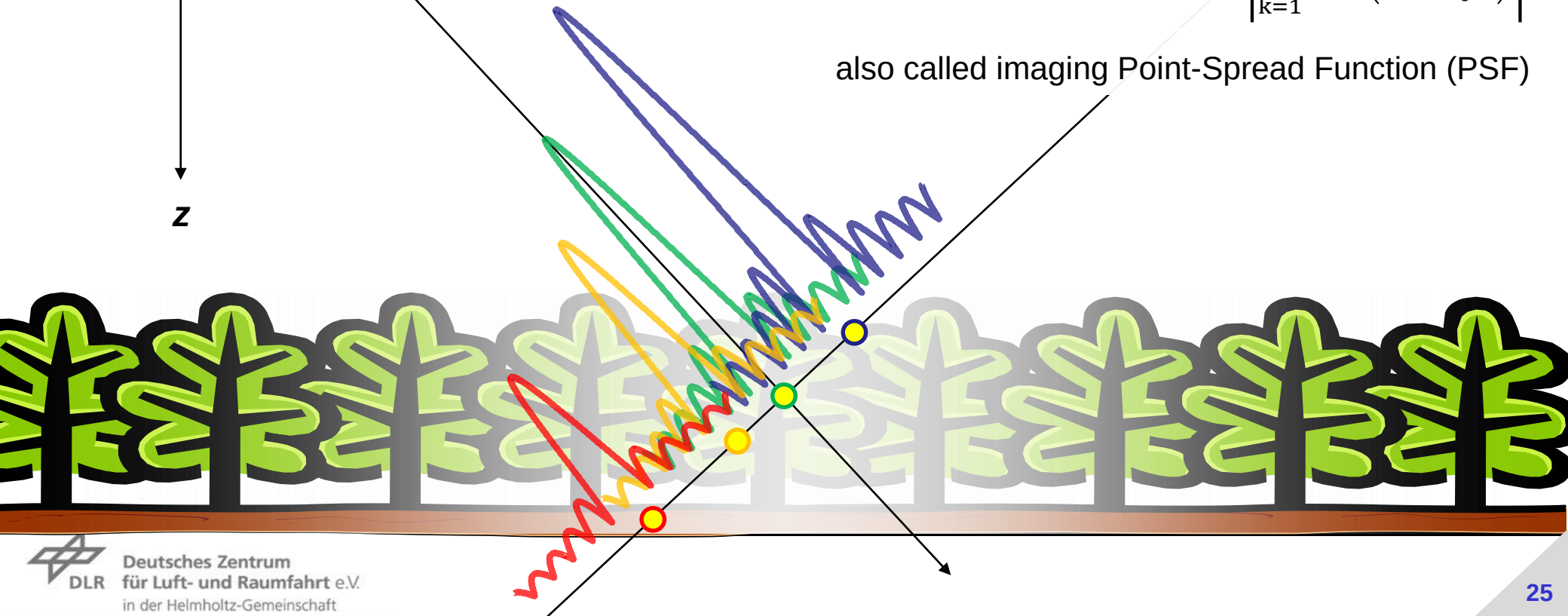
Normal aperture required for a **height resolution** δ_n : $L_x \geq \frac{\lambda}{2\delta_n} R_0$

Number of acquisitions with **uniform** displacements : $K \geq \frac{L_x}{d} + 1$

Cancellation function:

$$\Sigma(n) = \frac{1}{K} \left| \sum_{k=1}^K \exp \left(j \frac{4\pi d_k}{\lambda R_0} n \right) \right|$$

also called imaging Point-Spread Function (PSF)



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